Placeholder

Abstract

Strong gravitational lensing of gravitational waves is an expected consequence of the propagation of compact binary signals through an inhomogeneous universe, and a small but non-zero fraction of events detected by ground based observatories should be multiply imaged by intervening galaxies or clusters [1][2][3][4]. Recent forecasts indicate that the first confident strongly lensed gravitational wave detection could occur during the fifth LIGO-Virgo-KAGRA observing run (O5)[4], while searches incorporating O1-O4a events forecast a substantial probability of reaching at least 3σ significance in O5[5]. At the same time, simulations tailored to the wide field capabilities of the Euclid telescope demonstrate that high resolution imaging can successfully indentify and localize the strongly lensed host galaxies of merging binary black holes, significantly narrowing down the gravitatioanl wave sky localization area [6]. Motivated by these results, this work develops a Bayesian framework in which a catalogued lens with resolved strong lensing information is used to increase the confidence of a candidate lensed GW pair. We derive the full resolved strong lensing evidence, reconstruct the binary black hole population for prior odds, gravitational wave only bayes factor and joint analysis of the evidence, producing a joint bayes factor. The remaining hypotheses describe a catalogued lens without resolved images, an uncatalogued lens and two independent unlensed binary black holes. The central forecast to be tested is that targeted (EM) follow-up can, for favourable resolved strong lensing systems, sharpen the lens prediction sufficiently to raise the strong lensing detection confidence toward 5σ.

Keywords: gravitational waves; gravitational lensing: strong; Bayesian inference; electromagnetic catalogues; binary black holes; multi-messenger astronomy.

3σ to 5σ is a jump of 4500 more confident, which relates to the ratio of Bayes' factors for GW to GW+EM being 4500 but not sure if this is the same for the posterior odds as this would then assume the prior odds to be the same for GW only and joint, which we dont know if they are

1 Introduction

The field of gravitational waves (GWs) astronomy has rapidly transitioned from the first detection just about a decade ag to a large population of hundreds of detected compact-binary mergers. So far, the LIGO-Virgo-KAGRA (LVK) collaboration has repoted neraly 400 confirmed GW signals from compact binary coalescences,providing a an increasingly large sample with which we can use to search for rare astrophysical phonomena.

One of such phenomenon is strong gravitational lensing of GWs, which is expected to occur when a GW signal passes through the gravitational potential of an intervening galaxy or cluster. Despite the growing number of detected GW events, no confident detection of strong lensing has yet been reported. Current forecasts suggest that strong lensed GW detections might be detected in the next observing run O5 [5].

Gravitational waves propagate along null geodesics and are therefore subject to gravitational lensing in the same geometric-optics sense as electromagnetic radiation [1][7]. For galaxy-scale lenses, a single compact-binary coalescence can produce multiple detectable images separated by minutes to months, depending on the lens potential and source geometry. To leading order, the images share the same intrinsic binary parameters while differing in arrival time, amplitude through the lensing magnification and an image-type-dependent phase contribution [2][8][9][10]. This repeated signal structure motivates searches for pairs or multiples of GW events that are mutually consistent under a common source hypothesis.

The central statistical difficulty is that an unlensed population can also produce event pairs with overlapping posterior distributions. A robust lensing claim must therefore compare a common-source model against a model in which the detections arise from independent binaries. Bayesian model selection provides a natural framework because it combines likelihood agreement, prior information, population expectations and parameter-space volume into a single evidence ratio [11][12][13][14]. Searches of LIGO-Virgo-KAGRA data have so far found no compelling population level evidence for strong lensing, reinforcing the need for careful prior odds and follow up tests [15][16][17].

A major source of additional information is the electromagnetic foreground. A galaxy or cluster capable of producing strong lensing may be visible in imaging or spectroscopic catalogues even when the BBH source itself has no detectable EM counterpart. The available EM information can range from simple measurements of the foreground galaxy to a resolved strong-lensing system containing arcs, multiple images or an Einstein ring from an unrelated background source. These two types of information constrain different parts of the lens model. The foreground galaxy light constrains the luminous structure and population properties of the deflector, while resolved lensed source light constrains the mapping produced by the lens mass distribution.

For this reason, the parameterisation adopted here separates the EM lens model into source light parameters, lens light parameters and lens mass parameters. This is preferable to grouping all EM quantities into a single generic lens vector because

it makes clear which parameter blocks are directly measured and which are only inferred indirectly under each hypothesis.

2 Methodology

2.1 Simulated GW events and Bayesian parameter estimation

Do I have to elaborate on how we simulated the GW events, or briefly mention it and focus on how we use it?

2.2 Population reconstruction

The population distribution of binary black hole (BBH) parameters plays a crucial role in determining whether two similar gravitational wave (GW) signals originate from the same source or from two independent mergers exhibiting very similar properties. In densely populated regions of the parameter space, two unrelated GW events are more likely to exhibit similar inferred parameters purely by coincidence. Conversely, agreement between parameters that are rare in the underlying BBH population provides stronger evidence in favour of the lensing hypothesis.
To account for this population dependence in the Bayesian analysis, we reconstruct the BBH population distribution from simulated GW observations. We first reconstruct the observed population using posterior samples from detected events, then correct for selection effects to recover the underlying astrophysical population. The reconstructed population is subsequently used to generate a strongly lensed BBH population for prior odds estimation and Bayes factor computation.

2.2.1 Reconstruction of observed population

Population reconstruction on a simulated distribution of BBHs is performed using FIGARO, which uses a non-parametric approach to model probability densities using a Dirichlet Process Gaussian Mixture Model (DPGMM). The probability density is represented as a mixture of Gaussian components, with the number of components inferred from the data rather than specified in advance. Since the parameters of individual GW events are inferred with measurement uncertainties, we employ the hierarchical DPGMM (HDPGMM) framework to combine information from individual-event posterior distributions and reconstruct the underlying population density [18].
The reconstruction is performed on the joint three-dimensional distribution of the component masses m_1, m_2, and redshift z. For each event, the posterior density is first reconstructed using DPGMM, with multiple density realizations generated to characterize the uncertainty in the density estimate. These individual-event density reconstructions are subsequently combined hierarchically using HDPGMM to infer the population distribution (see Fig. 4, left panel, for the reconstructed population).

2.2.2 Correction for selection effects

The reconstructed observed population does not reflect the intrinsic population, as the reconstruction is based on posterior samples of detected events. In order to eliminate

selection effects and create an astrophysical population, we perform rejection sampling during generation of samples from the reconstructed observational population, with acceptance probability approximated to be a function of chirp mass.
The astrophysical BBH merger rate density is assumed to follow a redshift-dependent model,

$$\frac{R_{\rm GW}(z)}{\rm Gpc^{-3}yr^{-1}} = \frac{a_1 e^{a_2 z}}{e^{a_3 z} + a_4},$$

where $a_1 = 6600$, $a_2 = 1.6$, $a_3 = 2.1$, and $a_4 = 30$ for population-I/II stars[2].
The number of detectable events for a given chirp mass is given by

$$N_{\rm detected}(M_c) = \int_0^{z_{\rm horizon}(M_c)} \frac{dV_c}{dz} \frac{R(z)}{1+z} dz,$$

where $\frac{dV_c}{dz}$ is the differential comoving volume element and the factor (1/(1+z)) accounts for cosmological time dilation.
In importance sampling, the importance ratio, or the weight, is given by

$$w(x) \propto \frac{p(x)}{q(x)},\ x_i \sim q(x),$$

where $p(x)$ and $q(x)$ are the target population and biased population respectively.
In our case, it can written as

$$w(M_c) \propto \frac{P_{\rm intrinsic}(M_c)}{P_{\rm FIGARO}(M_c)},\ M_c \sim P_{\rm FIGARO}(M_c),$$

which is equivalent to

$$w(M_c) \propto \frac{P(M_c)}{P(M_c|\text{detected})}.$$

Applying Bayes theorem,

$$P(M_c|\text{detected}) = \frac{P(\text{detected}|M_c)P(M_c)}{P(\text{detected})},$$

where $P(\text{detected})$, the probability of detection, is independent of chirp mass, and

$$P(\text{detected}|M_c) = \frac{N_{\rm detected}(M_c)}{N_{\rm all}}.$$

Hence, the reweighting factor can be written as

$$w(M_c) \propto \frac{1}{N_{\rm detected}(M_c)},$$

which is then converted to acceptance probability $a(M_c)$ afterwards for rejection sampling:

$$a(M_c) = \frac{w(M_c)}{w_{\max}}.$$

Rejection sampling is then performed on draws from the observed population to obtain an approximately selection-corrected astrophysical population, which is subsequently used to compute the prior odds and Bayes factors.

The correction generally favours lower-mass systems, which are detectable over smaller cosmological volumes. This trend is important to capture when imposing strong lensing on the reconstructed population (see Fig. 4, right panel, for the reconstructed population).

2.2.3 Reconstruction of the strongly lensed population

To reconstruct the strongly lensed population, we use the LVK (LIGO–Virgo–KAGRA) Event Rate calculator and simulator (LeR) [19]. Component masses are drawn from the reconstructed BBH distribution corrected for selection effects, while source redshifts are sampled according to the astrophysical merger-rate distribution. The remaining GW source parameters are sampled from their respective distributions. For each source, a foreground lens galaxy is generated using LeR, with lens redshift, velocity dispersion, ellipticity, orientation, and external shear sampled according to the lens population model. The lens equation is then solved to obtain the image positions, magnifications, and arrival-time delays. The resulting lensing parameters are stored for subsequent analysis, and detectable lensed events are selected using LeR's GW detection-probability calculation.

The number of lensed images depends on the source position relative to the caustics, which are boundaries in the source plane separating regions with different image multiplicities. Sources located between the outer and inner caustics generally produce two images, while those inside the inner diamond-shaped caustic produce four images. For sufficiently elliptical lenses, the diamond caustic may extend beyond the outer caustic, allowing three-image configurations. To allow further investigation and comparison between different cases, we reconstruct the population for 2-image, 3-image, and 4-image cases separately. For each image multiplicity, we reconstruct the joint distribution of relative effective luminosity distances and arrival-time delays using FIGARO's DPGMM (see Fig. 6, 7, and 8 for the reconstructed lensed population).

2.3 Prior odds estimation

The prior odds are defined as the ratio of the probability that a given set of GW events consists entirely of strongly lensed images of the same source to the probability that this is not the case, rather than the ratio of the probability that an individual event is lensed to the probability that it is not [19]. As we have already corrected for selection effects in the population reconstruction, the prior odds are estimated using the detectable merger rates obtained from the reweighted population:

$$P_U^L = \frac{P(H_L)}{P(H_U)} \approx P(H_L) \approx N_{\text{img}}! \frac{\langle N_L \rangle}{\langle N \rangle^{N_{\text{img}}}},$$

where $N_{\rm img}$ is the number of images, $\langle N_L \rangle = R_L T$ is the expected number of detectable lensed systems, and $\langle N \rangle = (R_U + N_{\rm img} R_L)T$ is the expected total number of detected GW signals over an observation time T. Here, R_L and R_U denote the detectable lensed-system and unlensed-event rates, respectively [19].
The detectable rates are estimated using LeR from the simulated unlensed and lensed populations described in Section 2.2. To distinguish different image multiplicities, the detectable lensed systems are classified into 2-image, 3-image, and 4-image configurations. For each configuration, we require that all generated images satisfy the detection criterion. The rate corresponding to each configuration is estimated by weighting the total detectable lensed rate by the fraction of systems satisfying both the image-multiplicity and detectability requirements,

$$R_{L,n} = R_L \frac{N_{L,n}^{\rm all\,det}}{N_L^{\rm det}},$$

and systems for which only a subset of the generated images is detectable are excluded from the multiplicity-specific rates.
The prior odds for each image multiplicity are then calculated as

$$P_{U,n}^{L} \approx n! \frac{R_{L,n} T}{[(R_U + n R_{L,n})T]^n}, \qquad n = 2, 3, 4.$$

Since the lensed-event rate is much smaller than the unlensed-event rate, $R_{L,n} \ll R_U$, this can be approximated as

$$P_{U,n}^{L} \approx n! \frac{R_{L,n}}{R_U^n T^{n-1}}.$$

We adopt an observation time of $T = 1, \mathrm{yr}$ when evaluating the prior odds for the 2-, 3-, and 4-image cases.

2.4 Bayes factor computation

We adopt the PO2.0 framework developed by Barsode et al. [3] to compute the Bayes factor between the strongly lensed and unlensed hypotheses. The Bayes factor is defined as

$$B_U^L = \frac{P(d_1, \ldots, d_N \mid H_L)}{P(d_1, \ldots, d_N \mid H_U)},$$

where d_i denotes the GW data associated with the i-th event and N is the number of candidate images. PO2.0 evaluates the Bayes factor using posterior samples from individual-event parameter estimation, without requiring joint parameter estimation. The calculation incorporates the consistency of the intrinsic binary parameters and sky localization, together with lensing-induced differences in apparent luminosity distance, arrival time, and phase. We refer readers to Barsode et al. for the detailed formulation.
As suggested by Barsode et al., we generalize the original two-event PO2.0 implementation to accommodate N-image systems. The events are ordered by arrival time, with

the earliest-arriving image taken as the reference. The posterior samples of the reference image are transformed according to the relative magnifications and Morse phase differences, and compared with the posterior distributions of the remaining $N-1$ images. The calculation is extended to include the joint distributions of the relative lensing parameters, and contributions from different Morse phase configurations are combined according to their relative probabilities in the simulated lensed population. Following Equation B15 of Barsode et al. [3], we generalize the time-delay likelihood ratio to N candidate images. Under the unlensed hypothesis, the arrival times are assumed to be independently and uniformly distributed within an observation period $T_{\rm obs}$. For N events ordered by arrival time, $t_1 < t_2 < \cdots < t_N$, the joint probability density is

$$P(t_1, \ldots, t_N \mid H_U) = \frac{N!}{T_{\rm obs}^N}.$$

Defining the relative time delays as $\Delta t_j = t_j - t_1$, where $j = 2, \ldots, N$, and marginalizing over the first arrival time $0 < t_1 < T_{\rm obs} - \Delta t_N$, we obtain

$$P(\Delta t_2, \ldots, \Delta t_N \mid H_U) = \frac{N!(T_{\rm obs} - \Delta t_N)}{T_{\rm obs}^N}.$$

Since the reconstruction is performed using logarithmic time delays, we define the vector $\boldsymbol{x} = (x_2, \ldots, x_N)$, where $x_j = \log_{10}(\Delta t_j / \mathrm{s})$. Applying the corresponding Jacobian, we obtain

$$P(\boldsymbol{x} \mid H_U) = \frac{N!(T_{\rm obs} - \Delta t_N)}{T_{\rm obs}^N} (\ln 10)^{N-1} \prod_{j=2}^{N} \Delta t_j.$$

The corresponding joint distribution under the lensed hypothesis is estimated using kernel density estimation from the simulated lensed population. The time-delay likelihood ratio is therefore

$$\mathcal{R}_{LU}^{(N)} = \frac{P(\boldsymbol{x} \mid H_L)}{P(\boldsymbol{x} \mid H_U)}.$$

We adopt $T_{\rm obs} = 1\,\mathrm{yr}$, consistent with the observation time used for the prior-odds calculation. Finally, we automate the calculation over the simulated catalogues of 2-, 3-, and 4-image systems. For each system, the individual-event posterior samples and corresponding PE priors are loaded, preliminary posterior-overlap checks are performed, and the lensed and unlensed evidences are evaluated. For numerical stability, the logarithmic Bayes factors are restricted to the range $[-30, 30]$, with systems failing the preliminary posterior-overlap checks assigned $\log_{10} B_U^L = -30$. The resulting Bayes factors are combined with the prior odds calculated in Section 2.3 to obtain the posterior odds.

3 Bayesian hypothesis space

Consider two GW detections, $d_{GW} = d_{GW,1}, d_{GW,2}$, together with EM observations dEM drawin from a catalogue of potential foreground lenses. We divide the lensing interpretation into three mutually exclusive observational cases and compare their union with an unlensed alternative.

Table 1 Mutually exclusive hypotheses used in the joint GW-EM model comparison

Hypothesis	GW interpretation	EM/catalogue information
H_1	Two detections are lensed images of the same BBH.	The repsonsible lens is catalogues and resolved strong-lensing EM information constrains a detailed lens model.
H_2	Two detections are lensed images of the same BBH.	The responsible lens is catalogues, but no resolved strong-lensing image configuration is available; EM data constrain galaxy properties.
H_3	Two detections are lensed images of the same BBH.	The responsible foreground lens is absent from the catalogue.
H_U	The detections arise from two independent unlensed BBHs.	Any galaxy in the search region is treated as an unrelated field object, conditional on the search selection.

$$H_L = H_1 \cup H_2 \cup H_3 \tag{1}$$

with H_1, H_2 and H_3 mutually exclusive within the assumed search space. The posterior probability of the total lensed hypothesis is therefore

$$p(H_L|d) = p(H_1|d) + p(H_2|d) + p(H_3|d) \tag{2}$$

where $d = d_{GW}, d_{EM}$. For each model H_i, Bayes' theorem gives

$$p(H_i|d) = Z_i \frac{p(H_i)}{p(d)}, Z_i \equiv p(d|H_i). \tag{3}$$

The posterior odds of lensing relative to the unlsened hypothesis are consequently

$$\frac{p(H_L|d)}{p(H_U|d)} = \frac{[Z_1 p(H_1) + Z_2 p(H_2) + Z_3 p(H_3)]}{Z_U p(H_U)} \tag{4}$$

Eq. (4) makes the role of the hypothesis prior explicit. The Bayes factor quantifies how the data update the relative support for the models, while the prior odds encode the expected rarity of strong lensing and the probability that a responsible lens belongs to each catalogue class. This distinction is important because prior odds and search selection can change substantially as the GW catalogue grows [11][13][17].

4 Parameterisation and Conditional Structure

4.1 Binary Black Hole Source Parameters

Let θ_{BBH} denote the physical parameters of the compact binary source. A representative set is

$$\theta_{BBH} = \{m_1, m_2, \chi_1, \chi_2, z_s, \Omega_s, \iota, \phi_{pol}, ...\}. \tag{5}$$

Under any lensed hypothesis, the intrinsic source parameters are shared by the images, whereas under H_U the two detections have independent parameter sets $\theta_{BBH,1}$ and $\theta_{BBH,2}$. Apparent luminosity distances are image dependent because magnification rescales the observed strain amplitude.

4.2 Source light parameters

The source light block describes the resolved electromagnetic background source whose images or arcs are used to constrain the foreground lens. It does not represent an electromagnetic counterpart to the BBH. A generic source light parameter vector may be written

$$\theta_{SL} = \{\Omega_{s,EM}, F_s, R_{eff,s}, n_s, q_s, \phi_s, colour/SED_s, ...\} \tag{6}$$

Here $\Omega_{s,EM}$ is the unlensed source-plane centroid, F_s is the source flux or surface-brightness normalisation, $R_{eff,s}$ is an effective radius, n_s is a Sersic indec, q_s is the projected axis ratio and ψ_s is the source orientation. A pixelised or basis function source reconstruction may be used instead, in that case θ_{SL} represents the corresponding source light coefficients and regularisation hyperparameters.

The source light is required only when the resolved strong lensing EM data are modelled (under H_1). It is absent from H_2 because no resolved lensed source is availible, and absent from H_3 because no foreground lens candidate is indentified.

4.3 Lens light parameters

The lens light block describes the luminous distribution of the foreground deflector and the observables that determine whether it is present in the catalogue. A representative parameter vector is

$$\theta_{LL} = \{\Omega_l, F_l, R_{eff,l}, n_l, q_l, \psi_l, colour/SED_l, z_l, ...\} \tag{7}$$

These parameters describe the centroid, brightness, size, Sersic profile, axis ratio, position angle, spectral energy distribution information and redshift of the foregoround galaxy. Depending on the data set, the parameters may also contain velocity dispersion measurements or population level nuisance quantities associated with photomoetric redshift and stellar population inference.

Under H_1 and H_2, θ_{LL} is constrained directly by the catalogue or imaging data. Under H_3, θ_{LL} is not observed but can be retained as a latent population variable because catalogue completeness depends primarily on the brightness, size, redshift and surface brightness properties of the foregrouhd galaxy.

4.4 Lens mass parameters

The lens mass block describes the gravitational potential responsible for producing the multiple GW images. A model dependent vector may be written

$$\theta_{LM} = \{\theta_E \text{or} \sigma_v, \gamma^{'}, q_m, \phi_m, \Omega_m, \gamma_{ext}, \phi_{ext}, \kappa_{ext}, ...\} \tag{8}$$

Here $\theta_E \text{or} \sigma_v$ sets the mass scale, $\gamma^{'}$ is a density slope parameter where relevant, $q_m \text{and} \phi_m$ describe a projected mass ellipticity and orientation, Ω_m is the mass centroid, and $\gamma_{ext}, \phi_{ext} \text{and} \kappa_{ext}$ describe line of sight or enviornmental perturbations. The precise contents depend on whether the adopted model is, for example, a singular isothermal ellipsoid, an elliptical power law mass distribution or a more flexible composite model.

The lens mass parameters are the central bridge between GW and EM analyses. The GW pair constrains combination of θ_{LM} through the image relations, while the EM data constrain θ_{LM} either directly through resolved strong lensing morphology under H_1 or indirectly through tis relation to θ_{LL} under H_2.

4.5 Relation between the parameter blocks

The lens model parameter space is therefore organised as

$$\Theta_{lensmodel} = \{\theta_{SL}, \theta_{LL}, \theta_{LM}, \nu_{EM}\} \tag{9}$$

with the most important caveat that the available blocks depend on the hypothesis. The lens light and lens mass blocks need not be statistically independent. A physically motivated population prior may instead take the form

$$\pi(\theta_{LL}, \theta_{LM}) = \pi(\theta_{LM}|\theta_{LL})\pi(\theta_{LL}) \tag{10}$$

or the reverse conditioning, depending on how stellar light, velocity dispersion, halo mass and mass to light relations are modelled. Treating this correlation explicitly is especially important under H_2, where the lens light supplies the principle EM route to constraining lens mass.

4.6 Image Level Lensing Observables

The GW pair constrains a lower dimensional set of relations between the images,

$$\psi_{GW} = \{\Delta t, \mu_r, \Delta n, \Omega, ...\} \tag{11}$$

where Δt is the relative arrive time delay, μ_r is a relative magnification ratio, Δn denotes the relative Morse index, and Ω represents common sky localisation. These quantities are generated by the lens equation and Fermat potenital rather than being independent parameters,

$$\psi = f(\theta_{LM}, z_s, \beta; cosmology). \tag{12}$$

4.7 Electromagnetic Nuisance Parameters

EM observations require nuisance parameters ν_{EM} describing measurement and modelling uncertainties. Depending on the catalogue and imaging data, ν_{EM} may include the point spread function, photometruc calibration, specral energy distribution parameters, stellar mass to light ratio, background subtraction, photometric redshift calibration and other instrument or population level terms. These parameter should be maginalised rather than fixed when their uncertainty is non negligible, beause overly narrow EM constraints can artificially increase the GW-EM consistency factor.

5 Unlensed Hypothesis

Under H_U the detections correspond to two independent BBH mergers. The GW evidence is

$$Z_{GW,U} = \int p(d_{GW,1}|\theta_{BBH,1}, H_U)p(d_{GW,2}|\theta_{BBH,2}, H_U)\pi_U(\theta_{BBH,1}, \theta_{BBH,2})d\theta_{BBH,1}d\theta_{BBH,2} \tag{13}$$

The EM observations must not be allowed to depend pn whether the GW pair happens to be unlensed. We therefore introduce an EM field model, $H_{EM,field}$, that describes unrelated foreground galaxies under the same catalogue/search selection. When the GW and EM data are conditionally independent under H_U,

$$H_U \equiv H_{GW,U} \cap H_{EM,field} \tag{14}$$

$$Z_U = Z_{GW,U}Z_{EM,field}, Z_{EM,field} = p(d_{EM}|H_{EM,field}) \tag{15}$$

This term is essential whenever the lensed hypotheses recieve evidence from the existence or properties of a catalogue galaxy. Omitting the corresponding field galaxy probability from H_U would systematically favour catalogues lens hypotheses simply because a galaxy was searched for and found.

6 H_1: catalogued lens with resolved strong lensing information

6.1 Resolved EM data and parameter set

Under H_1, the two GW detections are strongly lensed images of the same BBH and the responsible foreground lens is present in the EM catalogue with resolved strong lensing information. (not sure about this but had a thought) The resolved EM source need not be the BBH counterpart; it provides an independednt probe of the same foreground gravitational potential. The EM data may therefore be separated schematically into measurements of the lens galaxy and measurements of the resolved strong lensing configuration.

$$d_{EM,c} = \{d_{LL,c}, d_{SL,c}\} \tag{16}$$

where $d_{LL,c}$ denotes measurements of the foreground lens light, while $d_{SL,c}$ denotes the resolved image positions, arc, an Einstein ring or pixel level surface brightness

data.

The parameter vector is

$$\Theta_1 = \{\theta_{BBH}, \beta_{GW}, \theta_{SL}, \theta_{LL}, \theta_{LM}, \nu_{EM}\} \tag{17}$$

The central coupling is θ_{LM}. The GW likelihood depends on θ_{LM} through the relative image properties, while the resolved EM likelihood constrains θ_{LM} through the mapping between θ_{SL} and the observed lensed surface brightness. The lens light parameters θ_{LL} model the foreground luminous component and can also imform physcially motivated priors on θ_{LM}.

6.2 Full joint evidence and prior

$$p(d_{GW}, d_{EM,c}|\Theta_1, H_1) = p(d_{GW}|\theta_{BBH}, \theta_{LM}, \beta_{GW}, H_1) p(d_{EM,c}|\theta_{SL}, \theta_{LL}, \theta_{LM}, \nu_{EM}, H_1). \tag{18}$$

The full evidence for candidate c is

$$Z_{1,c} = \int p(d_{GW}|\theta_{BBH}, \theta_{LM}, \beta_{GW}, H_1) p(d_{EM,c}|\theta_{SL}, \theta_{LL}, \theta_{LM}, \nu_{EM}, H_1) \pi_1(\Theta_1) d\Theta_1. \tag{19}$$

A useful schematic prior factorisation is

$$\pi_1(\Theta_1) = \pi_{BBH} \pi_{\beta_{GW}} \pi_{SL} \pi(\theta_{LL}, \theta_{LM}) \pi_{\nu_{EM}} S_1. \tag{20}$$

The joint prior $\pi(\theta_{LL}, \theta_{LM})$ allows correlations between galaxy light and mass to be retained. S_1 denotes the required GW detection, catalogue and resolved lens selection conditioning. The evidence intergrated over both the scientifically interesting lens mass parameters and the source light, lens light and nuisance parameters required by the EM reconstruction.

6.3 GW observables and the EM predictive distribution

The resolved EM data determine a posterior for the lens mass after marginalising over source light, lens light and EM nuisance parameters:

$$p(\theta_{LM}|d_{EM,c}, H_1) \propto \int p(d_{EM,c}|\theta_{SL}, \theta_{LL}, \theta_{LM}, \nu_{EM}, H_1) \pi_1 d\theta_{SL} d\theta_{LL} d\nu_{EM} \tag{21}$$

Propagating this mass posterior through the lens equation gives the predictive distribution

$$p(\psi|d_{EM,c}, H_1) = \int p(\psi|\theta_{LM}, z_s, \beta_{GW}, H_1) p(\theta_{LM}, z_s, \beta_{GW}|d_{EM,c}, H_1) d\theta_{LM} dz_s d\beta_{GW} \tag{22}$$

The corresponding prior corrected consistency statistic is

$$C_1 = \int \frac{p(\psi|d_{GW}H_1)p(\psi|d_{EM,c},H_1)}{\pi(\psi|H_1)}d\psi \tag{23}$$

A large C_1 indicates that the resolved EM reconstruction predicts time delays, magnification relations and image properties compatible with the GW pair. A small value indicates tension. When the full evidence in Eq. (19) is evaluated directly, this consistency is already encoded through θ_{LM}; C_1 is therefore most safely interpreted as a reduced space diagnostic unless an explicit non overlapping factorisation is derived.

6.4 Multiple resolved lens candidates

If more than one resolved strong lensing system is compatiable with the GW localisation, the lens indentity should be marginalised rather than fixed in advance:

$$Z_1 = \sum_{c=1}^{N_{cat}} p(c|H_1)Z_{1,c} \tag{24}$$

7 Reduced GW-EM consistency under H_1

The reduced consistency statistic isolates the direct comparison between the GW inferred image relations and the prediction from the resolved EM lens reconstruction. Its prior denominator removes the common prior contribution that would otherwise be counted twice when multiplying posteriors.

In the present parameterisation, the key reaon H_1 can be highly constraining is that the resolved source light morphology θ_{SL} is lensed by the same θ_{LM} that generates the GW image relations. Marginalising over θ_{SL} and θ_{LL} can still leave a comparatively narrow posterior on θ_{LM}, provided the strong lensing configuration is informative.

8 Computational evaluation of H_1

The integrals can be manipulated and approximated using laplace to get a form more suitable for computational analysis. Using the joint maximum and laplace approximation the full evidence for H_1 can be written, in log form as,

$$\ln(Z_{1,c}) \approx \ell_1(\hat{\Theta}_1) + (\frac{k_1}{2})\ln(2\pi) - (\frac{1}{2})\ln(|A_1|) \tag{25}$$

Eq. (25) separates the evidence into a best fit contribution, $\ell_1(\hat{\Theta}_1)$, and a local posterior volume contribution determined by k_1 and $|A_1|$. The resolved EM data can alter both the peak term and the local posterior volume. In the source light/lens light/lens mass parameterisation, the full Hessian contains cross block curvature. In particular, degeneracies between source light and lens mass, and between lens light and lens mass, should be retained rather tha ignored by approximating the blocks independently.

9 Joint Bayes factor and posterior odds

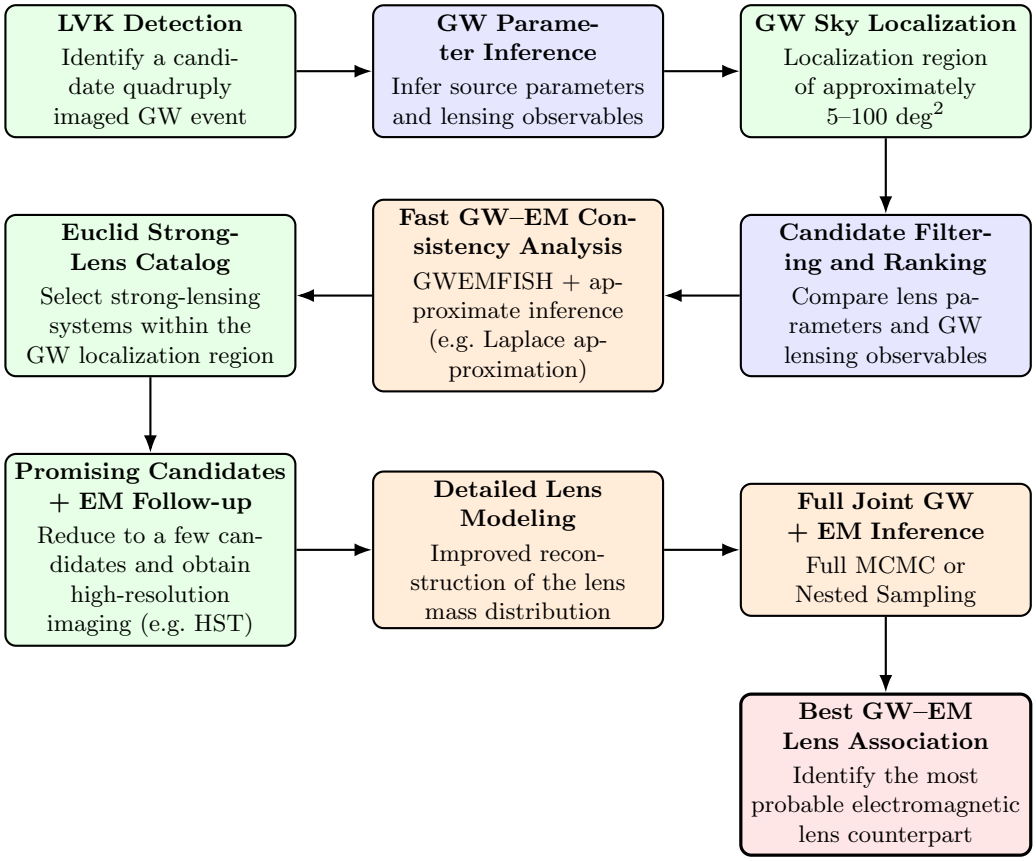


Figure 1 Conceptual workflow for identifying the electromagnetic lens counterpart of a quadruply imaged gravitational-wave event.

9.1 GWEMFISH and Laplace's Approximation: A physicist's life (and time) saver

Lens reconstruction will usually take very long (from hours to days) if it was ran on a traditional Markov Chain Monte Carlo program. To overcome this issue and ensure that the lens reconstruction can be computed quicky, we use GWEMFish, which utilizes Laplace approximation, to ensure that each computation is tubo-fast. -more details or any references to other sections on how this works

Using Laplace's approximation, we can approximate the evidence for each lens candidate in the GW localization region. This allows us to quickly compute the Bayes factor for each candidate and rank them based on their likelihood of being the correct lens. With this, we can much more easily narrow down the candidates and focus

on the most promising ones for further analysis and follow-up observations, such as performing a full MCMC or Nested Sampling and follow up using a telescope with higher resolution such as the HST.

9.2 Lens searches and the localization problem

It is known that GWs poor localization compared to other EM counterparts (normally around $5 - 25 deg^2$). In a region of sky of that size, it is predicted by (referece) that the Euclid telescope could detect up to around 600 lensing galaxies. This means that we can't immediately know which galaxy the GW signal originates. This is analogous to the birthday problem where we would need to perform searches with the lensing candidates in the localized region of sky to match the GW signal to the correct lens. Previous studies (refereces) have concluded that the correct lens would equate to being able to localize the GW correctly.

9.3 From an ominiscient point of view: the joint Bayes factor

10 Result

10.1 Reconstructed Population

10.2 Prior Odds

Table 2 Estimated detectable gravitational-wave event rates, lensed-to-unlensed rate ratios, and prior odds for 2-image, 3-image, and 4-image systems. The total detectable unlensed-event rate is $R_U = 448.76\,\mathrm{yr}^{-1}$, while the total detectable lensed-system rate is $R_L = 0.44\,\mathrm{yr}^{-1}$. The multiplicity-specific rates require all generated images to satisfy the GW detection criterion. The prior odds are evaluated for an observation time of $T = 1\,\mathrm{yr}$.

Quantity	2 images	3 images	4 images
$R_{L,n}\,(\mathrm{yr}^{-1})$	0.19	0.02	0.07
$R_{L,n}/R_U$	4.2×10^{-4}	5.0×10^{-5}	1.5×10^{-4}
$P^L_{U,n}$	1.9×10^{-6}	1.5×10^{-9}	3.9×10^{-11}

10.3 Bayes Factor and Comparison

11 Conclusion

References

[1] Li, S.-S., Mao, S., Zhao, Y., Lu, Y.: Gravitational lensing of gravitatioanl waves: a statistical perspective. Monthly Notices of the Royal Astronomical Society **476**, 2220–2229 (2018)

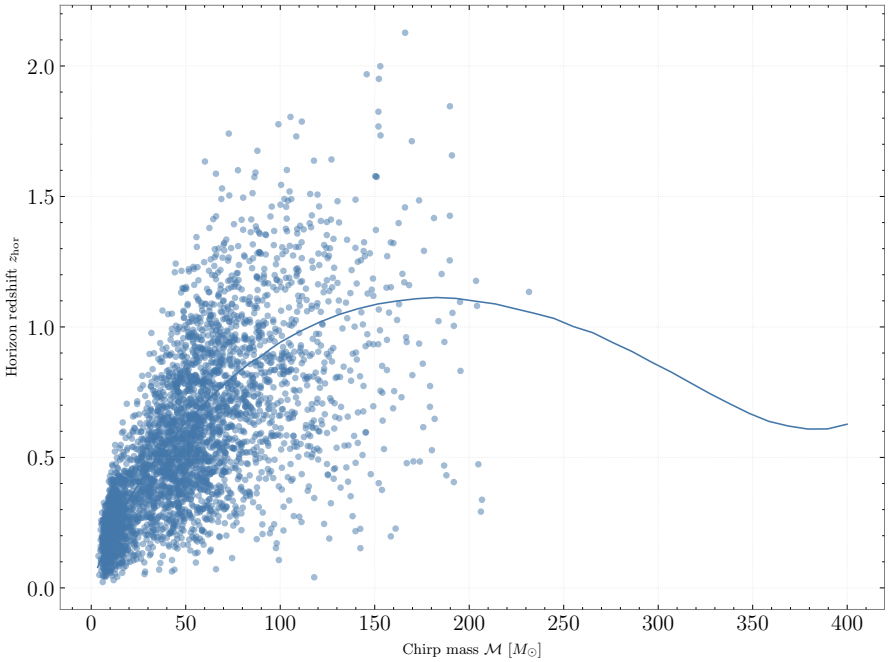


Figure 2 Horizon redshift $z_{\rm horizon}$ as a function of chirp mass $\mathcal{M}_c$. The scattered points represent the simulated detectable binary black hole events, while the solid curve shows the estimated horizon redshift for each chirp mass. The horizon redshift increases with chirp mass up to approximately 180 $M_\odot$, before decreasing at higher masses, reflecting the mass-dependent sensitivity of gravitational-wave detectors.

[2] Oguri, M.: Effect of gravitational lensing on the distribution of gravitational waves from distant binary black hole mergers. Monthly Notices of the Royal Astronomical Society **480**, 3842–3855 (2018)

[3] Barsode, A., Goyal, S., Ajith, P.: Fast and efficient bayesian method to search for strongly lensed gravitational waves. Astrophysical Journal **980**, 258 (2025) https://doi.org/10.3847/1538-4357/adae10

[4] Barsode, A., Maity, K.N., Ajith, P.: Lensing, not luck! detection prospects of strongly lensed gravitational waves. Astrophysical Journal **998**, 294 (2026) https://doi.org/10.3847/1538-4357/ae38c1

[5] Barsode, A., Maity, K.N., Ajith, P.: Search for strong lensing of gravitatioanl waves int he binary black hole events from o1-o4a. arXiv preprint (2026) arXiv:2607.08466

[6] Wempe, E., Koopmans, L.V.E., Wierda, A.R.A.C., Hannuksela, O.T., Broeck, C.V.D.: On the detection and precise localization of merging black hole events through strong gravitational lensing. arXiv preprint (2024) arXiv:2204.08732

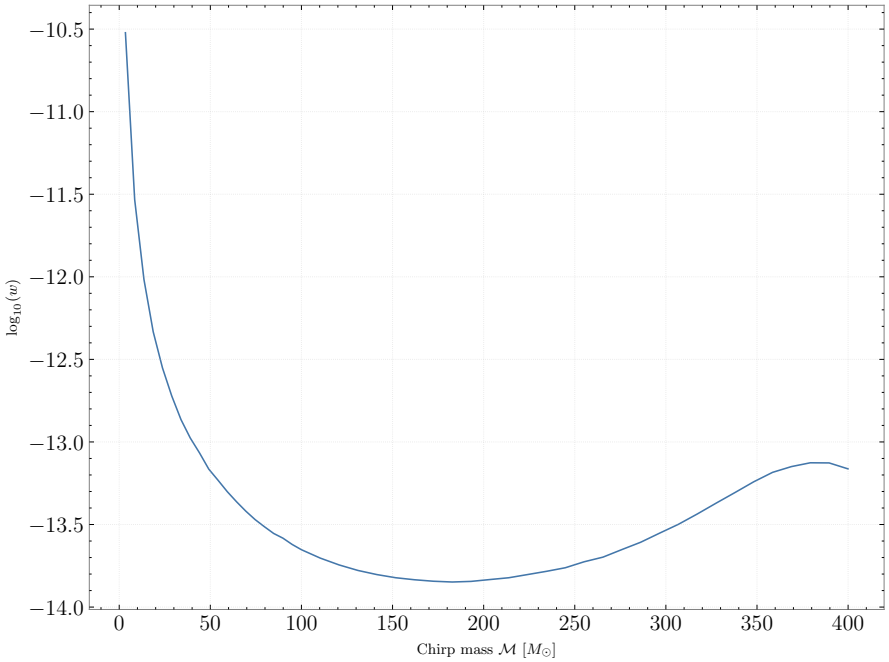


Figure 3 Logarithm of the selection-correction weight as a function of chirp mass $\mathcal{M}_c$, calculated from the expected number of detectable mergers within the horizon redshift. The weight decreases sharply with increasing chirp mass at lower masses, reaching a minimum around 180–200 $M_\odot$, before increasing gradually at higher masses. Larger weights correspond to systems with lower detection efficiency and therefore a greater probability of being retained during the rejection sampling used to reconstruct the astrophysical population.

[7] Takahashi, R., Nakamura, T.: Wave effects in gravitational lensing of gravitational waves from chirping binaries. Astrophysical Journal **595**, 1039–1051 (2003)

[8] Haris, K., Mehta, A.K., Kumar, S., Venumadhav, T., Ajith, P.: Identifying strongly lensed gravitational wave signals from binary black hole mergers. arXiv preprint (2018) arXiv:1807.07062

[9] Hannuksela, O.A., al.: Search for gravitational lensing signatures in ligo-virgo binary black hole events. Astrophysical Journal Letters **874**, 2 (2019)

[10] Ezquiaga, J.M., Holz, D.E., Hu, W., Lagos, M., Wald, R.M.: Phase effects from from strong gravitational lesning of gravitatioanl waves. Physics Review D **103**, 064047 (2021)

[11] Lo, R.K.L., Hernandez, I.M.: Bayesian statistical framework for indentifying strongly lensed gravitatioanl wave signals. Physical Review D **103**, 123015 (2021)

[12] Janquart, J., Hannuksela, O.A., Haris, K., Broeck, C.V.D.: Golum: a fast and precise methodology to search for, and analyze, strongly lensed gravitatioanl wave

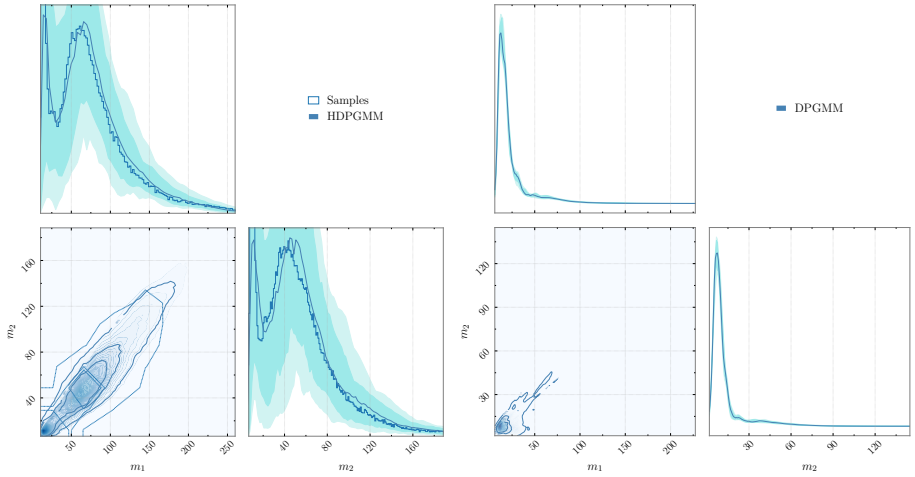


Figure 4 Reconstructed observed (left panel) and astrophysical (right panel) binary black hole population in component masses m_1 and m_2 using the hierarchical DPGMM implementation in FIGARO inferred from simulated population of gravitational-wave signals mimicking observing run O4. The diagonal panels show the one-dimensional marginal distributions, while the off-diagonal panels show the corresponding two-dimensional joint distributions. The observed population is biased towards higher masses due to selection effects, while the astrophysical population is corrected for these effects and represents the true underlying distribution of binary black hole mergers, extending to higher redshifts and lower masses that are relevant to strong gravitational-wave lensing, which occurs for binaries at high redshift.

events. Monthly Notices of the Astronomical Society **515**, 3989–3996 (2022)

[13] Janquart, J., Haris, K., Hannuksela, O.A., Broeck, C.V.D.: The return of golum: improving distributed joint parameter estimation for strongly lensed gravitational waves. Monthly Notices of the Royal Astronomical Society **526**, 3088–3098 (2023)

[14] Janquart, J., al.: Follow up analyses to the o3 ligo-virgo-kagra lensing searches. Monthly Notices of the Royal Astronomical Society **526**, 3832–3860 (2023)

[15] LIGO Scientific Collaboration, Virgo Collaboration: Search for lensing signatures in the gravitational wave observations from the first half of ligo-virgo's third observing run. Astrophysical Journal **923**, 14 (2021)

[16] LIGO Scientific Collaboration, Virgo Collaboration, KAGRA Collaboration: Search for gravitational lensing signatures in the full third observing run of the ligo-virgo network. Astrophysical Journal **970**, 191 (2024)

[17] LIGO Scientific Collaboration, Virgo Collaboration, KAGRA Collaboration: Gwtc-4.0: Searches for gravitational wave lesning signatures. arXiv preprint (2025) arXiv:2512.16347

[18] Rinaldi, S., Pozzo, W.D.: (H)DPGMM: a hierarchy of Dirichlet process Gaussian mixture models for the inference of the black hole mass function. Monthly Notices

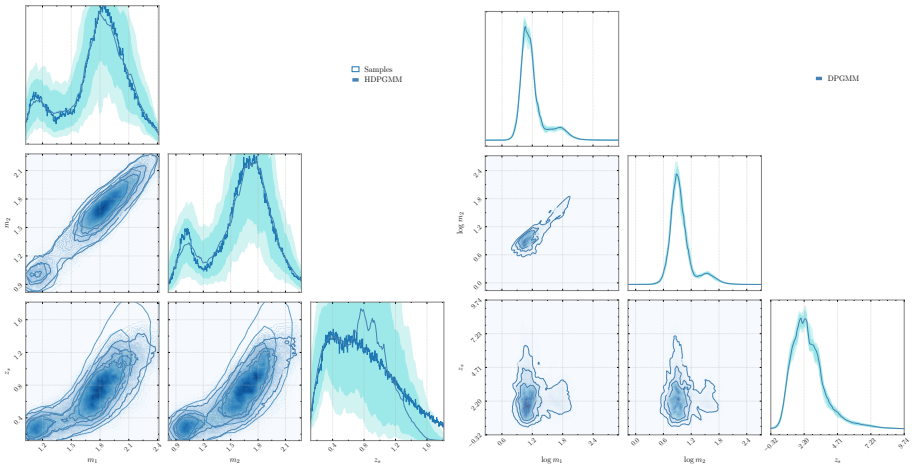


Figure 5 Reconstructed observed (left panel) and astrophysical (right panel) binary black hole population in log10 of component masses m_1 and m_2 using the hierarchical DPGMM implementation in FIGARO inferred from simulated population of gravitational-wave signals mimicking observing run O4. The diagonal panels show the one-dimensional marginal distributions, while the off-diagonal panels show the corresponding two-dimensional joint distributions. The observed population is biased towards higher masses due to selection effects, while the astrophysical population is corrected for these effects and represents the true underlying distribution of binary black hole mergers, extending to higher redshifts and lower masses that are relevant to strong gravitational-wave lensing, which occurs for binaries at high redshift.

of the Royal Astronomical Society **509**(4), 5454–5466 (2022) https://doi.org/10.1093/mnras/stab3224

[19] Hannuksela, O.A., Haris, K., Janquart, J., Narola, H., Phurailatpam, H., Creighton, J.D.E., Broeck, C.V.D.: Strong gravitational-wave lensing posterior odds. Astrophysical Journal **1002**, 42 (2026) https://doi.org/10.3847/1538-4357/ae5816

[20] Haris, K., Mehta, A.K., Kumar, S., Venumadhav, T., Ajith, P.: Identifying strongly lensed gravitational wave signals from binar black hole mergers. arXiv preprint (2018) arXiv:1807.07062

[21] Janquart, J., Hannuksela, O.A., Haris, K., Broeck, C.V.D.: A fast and precise methodology to search fro and analyse strongly lensed gravitatioanl wave events. Monthly Notices of the Royal Astronomical Society **506**, 5430–5438 (2021)

[22] Mahalanobis, C.P.: On the generalised distance in statistics. Proceedings of the National Institue of Sciences of India **2**, 49–55 (1936)

[23] Fisher, A.R.: On the mathematical foundations of theoretical statistics. Philosophical Transactions of the Royal Society of London A **222**, 309–368 (1922)

[24] Vallisneri, M.: Use and abuse of the fisher information matrix in the assessment

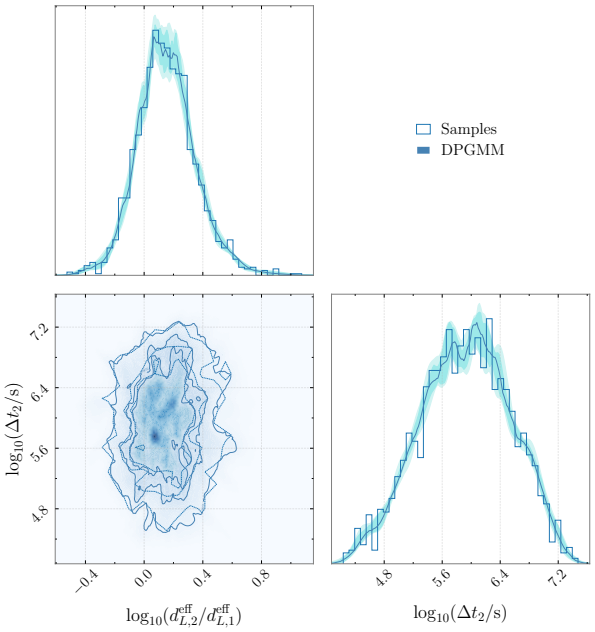


Figure 6 Reconstructed strongly lensed binary black hole population for 2-image systems using FIGARO's DPGMM. The reconstruction is performed on the joint distribution of the logarithmic relative effective luminosity distance $\log_{10}(d^{\mathrm{eff}}_{L,2}/d^{\mathrm{eff}}_{L,1})$ and logarithmic arrival-time delay $\log_{10}(\Delta t_2/\mathrm{s})$, with the earliest-arriving image taken as the reference. The diagonal panels show the one-dimensional marginal distributions, while the off-diagonal panel shows the corresponding two-dimensional joint distribution. The reconstructed density reproduces the main features of the simulated lensing-parameter distribution.

of gravitational wave parameter estimation prospects. Physical Review D **77**, 042001 (2008)

Appendix A H_2: catalogued lens without resolved strong lensing EM images

Available parameter blocks

H_2 retains the common source lensing interpretation but removes the resolved source light information. The foreground galaxy is catalogued and its lens light parametersare constrained by imaging, photometry and spectroscopy, but no resolved background source is available from which to reconstruct the lens mass directly.

$$\Theta_2 = \{\theta_{BBH}, \beta_{GW}, \theta_{LL}, \theta_{LM}, \nu_{EM}\} \tag{A1}$$

The absence of θ_{SL} is the defining parameter space difference from H_1. Lens mass is constrained only through the GW data and through the population or scaling relation connecting θ_{LL}toθ_{LM}.

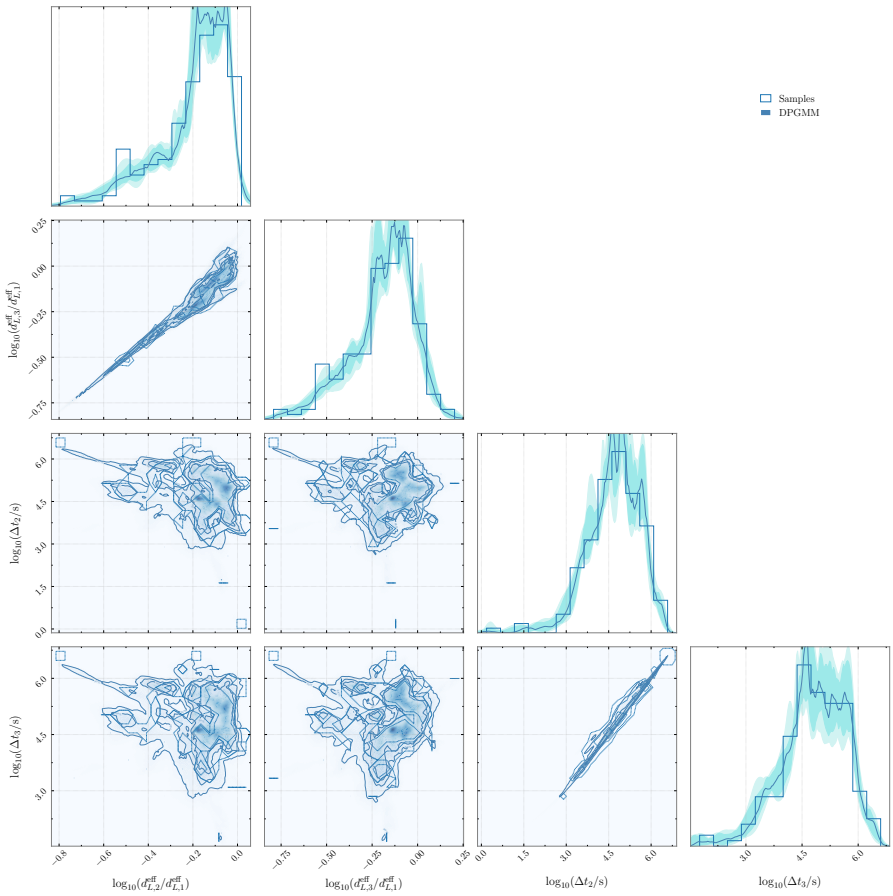


Figure 7 Reconstructed strongly lensed binary black hole population for 3-image systems using FIGARO's DPGMM. The reconstruction is performed on the joint four-dimensional distribution of logarithmic relative effective luminosity distances $\log_{10}(d^{\rm eff}_{L,j}/d^{\rm eff}_{L,1})$ and logarithmic arrival-time delays $\log_{10}(\Delta t_j/{\rm s})$, where $j = 2, 3$. The images are ordered by arrival time, with the earliest-arriving image taken as the reference. The diagonal panels show the one-dimensional marginal distributions, while the off-diagonal panels show the corresponding two-dimensional joint distributions. The reconstructed density captures the main features and correlations of the simulated lensing parameters.

Full evidence

$$p(d_{GW}, d_{EM,c}|\Theta_2, H_2) = p(d_{GW}|\theta_{BBH}, \theta_{LM}, \beta_{GW}, H_2)p(d_{EM,c}|\theta_{LL}, \nu_{EM}, H_2) \quad \text{(A2)}$$

Because the catalogue likelihood primarily constrains θ_{LL}, the connection to θ_{LM} is carried by the joint prior $\pi(\theta_{LL}, \theta_{LM})$, rather than by a resolved lensing likelihood. The full evidence is

$$Z_{2,c} = \int p(d_{GW}|\theta_{BBH}, \theta_{LM}, \beta_{GW}, H_2)p(d_{EM,c}|\theta_{LL}, \nu_{EM}, H_2)\pi_2(\Theta_2)d\Theta_2 \quad \text{(A3)}$$

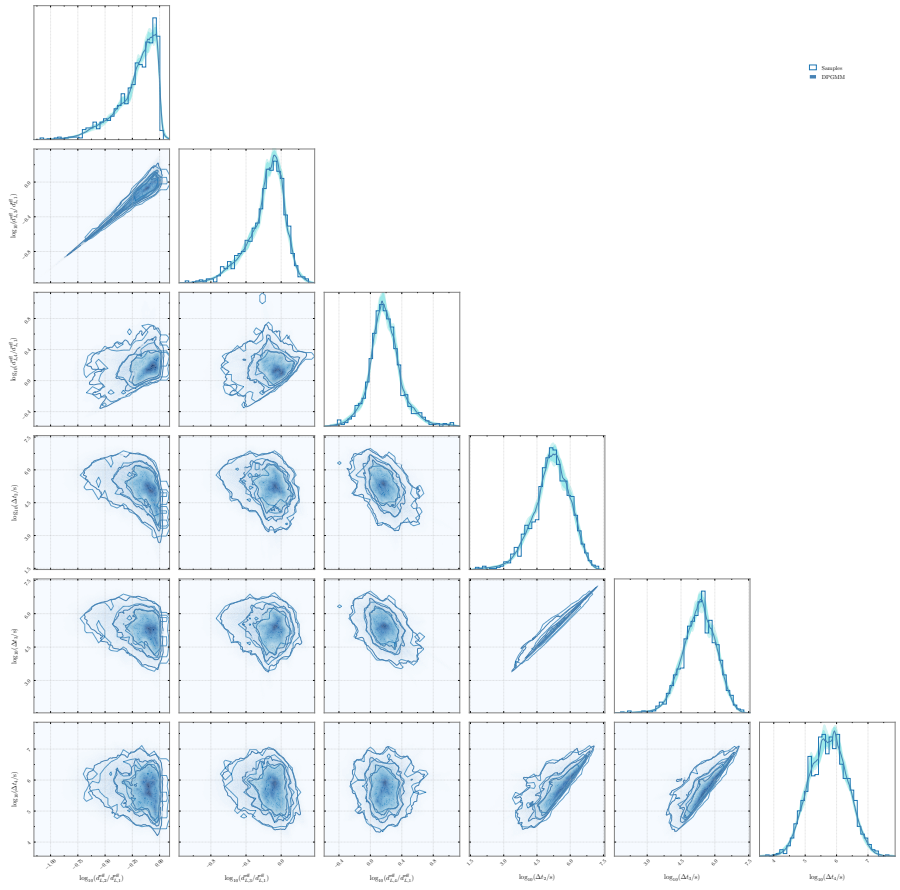


Figure 8 Reconstructed strongly lensed binary black hole population for 4-image systems using FIGARO's DPGMM. The reconstruction is performed on the joint six-dimensional distribution of logarithmic relative effective luminosity distances $\log_{10}(d^{\rm eff}_{L,j}/d^{\rm eff}_{L,1})$ and logarithmic arrival-time delays $\log_{10}(\Delta t_j/{\rm s})$, where $j = 2, 3, 4$. The images are ordered by arrival time, with the earliest-arriving image taken as the reference. The diagonal panels show the one-dimensional marginal distributions, while the off-diagonal panels show the corresponding two-dimensional joint distributions. The reconstructed density captures the main features and correlations of the simulated lensing parameters.

with a schematic prior

$$\pi_2(\Theta_2) = \pi_{BBH}\pi_{\beta_{GW}}\pi(\theta_{LL}, \theta_{LM}\pi_{\nu_{EM}}S_2 \tag{A4}$$

This form is preferable to treating the EM likelihood as if it measured θ_{LM} directly. Any inference from luminosity, effective radius, redshift or velocity dispersion to lens mass should enter through the appropriate conditional population model.

EM induced lens mass and image observable distributions

The catalogue information induces a lens mass posterior by marginalising over the

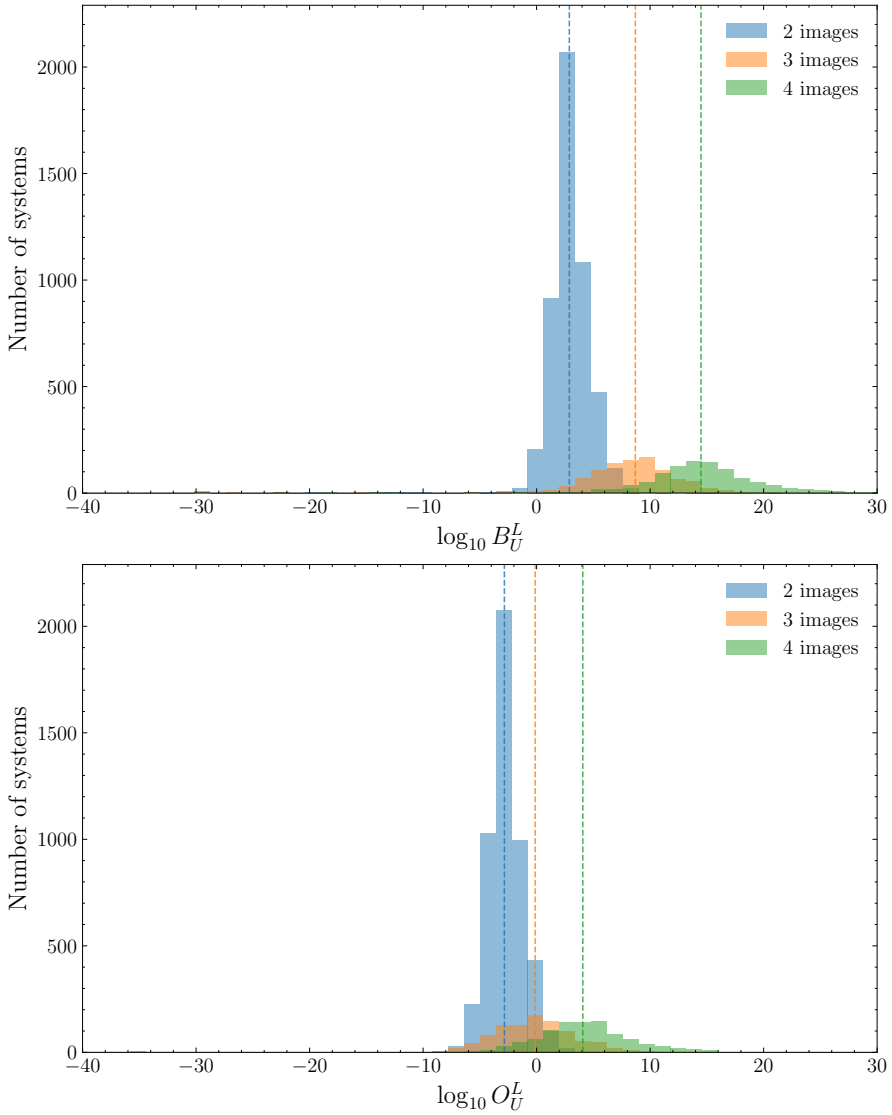


Figure 9 Distributions of the logarithmic Bayes factors $\log_{10} B_U^L$ (upper panel) and posterior odds $\log_{10} O_U^L$ (lower panel) for simulated strongly lensed 2-image (blue), 3-image (orange), and 4-image (green) systems. The dashed vertical lines indicate the corresponding distribution peaks. The Bayes factor distributions shift towards larger values with increasing image multiplicity, reflecting the additional information provided by multiple lensed images. After incorporating the prior odds, the posterior odds distributions shift towards lower values due to the low astrophysical probability of strong lensing.

measured lens light parameters:

$$p(\theta_{LM}|d_{EM,c}, H_2) \propto \int p(d_{EM,c}|\theta_{LL}, \nu_{EM}, H_2)\pi(\theta_{LL}, \theta_{LM})\pi_{\nu_{EM}} d\theta_{LL} d\nu_{EM} \quad \text{(A5)}$$

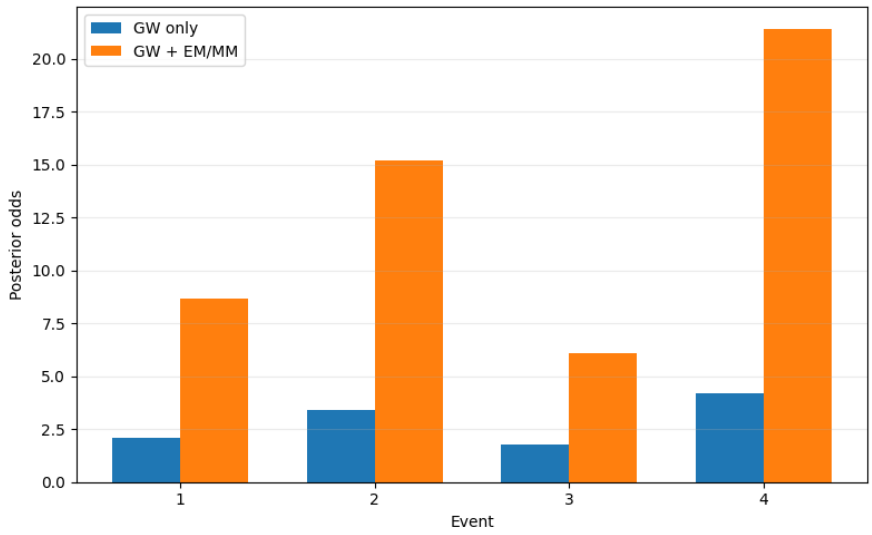


Figure 10

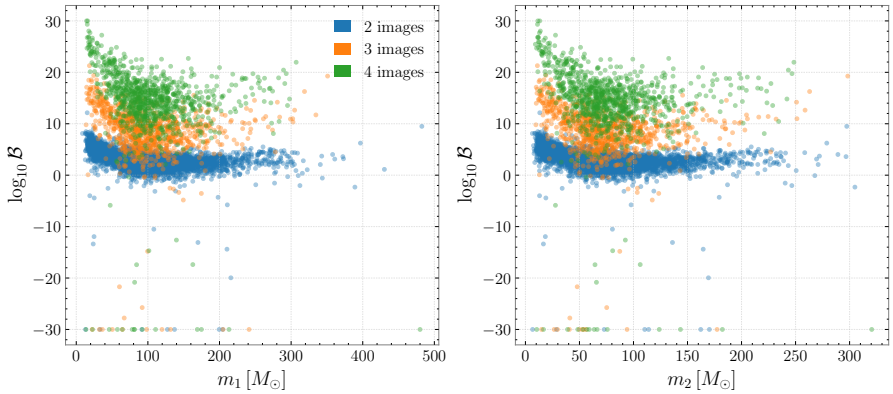


Figure 11 Logarithmic Bayes factors $\log_{10} B_U^L$ as functions of component masses m_1 (left panel) and m_2 (right panel) for simulated strongly lensed 2-image (blue), 3-image (orange), and 4-image (green) systems. The Bayes factors are generally larger for systems with higher image multiplicities, with the highest values predominantly occurring at lower component masses. The distributions show considerable scatter and overlap between image multiplicities. Values at $\log_{10} B_U^L = -30$ correspond to the imposed numerical cutoff, including systems failing the preliminary posterior-overlap checks.

This distribution can then be propagated to the GW image observables:

$$p(\psi|d_{EM,c}, H_2) = \int p(\psi|\theta_{LM}, z_s, \beta_{GW}, H_2) p(\theta_{LM}, z_s, \beta_{GW}|d_{EM,c}, H_2) d\theta_{LM} dz_s d\beta_{GW} \tag{A6}$$

Relative to H_1, this disribution is generally broader because θ_{LM} is inferred indirectly from lens light rather than from a resolved source light configuration.

Reduced consistency statistic

$$C_2 = \int \frac{p(\psi|d_{GW}, H_2)p(\psi|d_{EM,c}, H_2)}{\pi(\psi|H_2)} d\psi \tag{A7}$$

As under H_1, C_2 is best regarded as a diagnotstic when the full joint evidence is already evaluated, because the common lens mass information is then already included in Eq. (A3).

Laplace approximation

$$\ell_2(\Theta_2) = \ln(p(d_{GW}|\Theta_2, H_2)) + \ln(p(d_{EM,c}|\Theta_2, H_2)) + \ln(\pi_2(\Theta_2)) \tag{A8}$$

$$\hat{\Theta}_2 = \arg\max_{\Theta_2} \ell_2(\Theta_2) \tag{A9}$$

$$A_2 = -\nabla\nabla\ell_2(\Theta_2)|_{\hat{\Theta}_2} \tag{A10}$$

$$\ln(Z_{2,c}) \simeq \ell_2(\hat{\Theta}_2) + (\frac{k_2}{2})\ln(2\pi) - (\frac{1}{2}\ln(|A_2|) \tag{A11}$$

The curvature matrix A_2 contains cross information between θ_{LL}andθ_{LM} through their population relation. Because no θ_{SL} block is present, the EM data does not provide the strong image geometry constraints available under H_1. The resulting posterior in the lens mass directions will therefore normally be broader and more prior dependent.

Restriction of H_2

H_2 is restricted to systems in which the responsible lens is present in the catalogue but no resolved strong lensing EM configuration is used. If a resolved lensed source is available, the system belongs ot H_1, if the responsible foreground lens is absent from the catalogue, it belongs to H_3.

The principle physical limitation is that lens light does not uniquely determine lens mass. The relation $\pi(\theta_{LM}|\theta_{LL})$ must therefore include intrinsic scatter and relevent uncertainties. Treating the lens light measurement as a deteministic mass estimate would artificially narrow the H_2 evidence.

The source redishift and GW source plane position remain latent unless measure independetly. The Laplace approximation also requires an approximately single mode, locally Gaussian posterior. Weak lens mass constraints and light to mass degeneracies can make this assumption poor.

Appendix B H_3: uncatalogued lens

Latent lens light and lens mass population

Under H_3, the GW pair is lensed but the responsible foreground lens is not present in the catalogue. No condidate specific light source or lens light likelihood exists. Nevertheless, retaining a latent θ_{LL} bloack is usedful because catalogue detection depends on the luminous properties of the deflector, while the lensing signal depends on θ_{LM}.

$$\Theta_3 = \{\theta_{BBH}, \beta_{GW}, \theta_{LL}, \theta_{LM}\} \tag{B12}$$

The catalogue completeness function is written

$$C_{cat}(\theta_{LL}, \theta_{LM} = P(\text{lens appears in catalogue}|\theta_{LL}, \theta_{LM}) \tag{B13}$$

where dependence on θ_{LM} may be retained when survey selection is correlated with mass through velocity disperiosn, halo mass or enviromental properties. In the simplest implementation the completenedd may depend only on the observable lens light block.

Full evidence

The H_3 evidence becomes

$$Z_3 = \int p(d_{GW}|\theta_{BBH}, \theta_{LM}, \beta_{GW}, H_3)[1 - C_{cat}(\theta_{LL}, \theta_{LM})]\pi_{BBH}\pi_{\beta_{GW}}\pi(\theta_{LL}, \theta_{LM})d\Theta_3 \tag{B14}$$

This form makes the parameter space structure explicit. The GW likelihood selects lens mass configurations capable of producing the observed image relations, the joint lens light.lens mass population prior supplies physically plausiable deflectors, and the non-detection factor supresses systems whose light properties imply that they should have appeared in the catalogue.

Log integrand and laplace approximation

$$\ell_3(\Theta_3) = \ln(p(d_{GW}|\theta_{BBH}, \theta_{LM}, \beta_{GW}, H_3)) + \ln([1 - C_{cat}(\theta_{LL}, \theta_{LM})]) + \ln(\pi_{BBH}) + \ln(\pi_{\beta_{GW}}) + \ln(\pi(\theta_{LL}, \theta_{LM})) \tag{B15}$$

$$\hat{\Theta}_3 = \arg\max_{\Theta_3} \ell_3(\Theta_3) \tag{B16}$$

$$A_3 = -\nabla\nabla\ell_3(\Theta_3)|_{\hat{\Theta}_3} \tag{B17}$$

$$\ln(Z_3) \simeq \ell_3(\hat{\Theta}_3) + (\frac{k_3}{2})\ln(2\pi) - (\frac{1}{2})\ln(|A_3|) \tag{B18}$$

The catalogue non-detection term can alter both the height of the dominant mode and the curvature of the posterior. A lens mass solution that fits the GW well can still be suppressed if its correlated lens light properties make non-detection improbable.

Why there is no source light consistency factor under H_3

There is no condidate specific resolved background source under H_3, and therefore no surce light block θ_{SL} and no EM reconstruction from which to contrsuct

$p(\psi|d_{EM,c}, H_3)$. The role played by C_1andC_2 is replaced by marginalisation over the latent lens light/lens mass population and the catalogue non-detection probability.

Restriction of H_3

H_3 applies only when the responsible lens is absent from the adopted catalogue. The completeness model must therefore refer to exactly the same survery footprint, depth and object selection criteria used to define H_1andH_2.

The joint population model $\pi(\theta_{LL}, \theta_{LM})$ is essential. If lens light is omitted completely, the catalogue non-detection probability must be expressed indirectly as a function of mass, which can hide substantial astrophyscial uncertainty. Retaining θ_{LL} as a latent variable makes the selection model more transparent.

The laplace approximation can fail when catalogue completeness changes sharply near a survey threshold, when the light to mass population relation is strongly non-Gaussian or when several lens mass configurations provide comparable fits to the GW data.

Appendix C Laplace approximation to the full H_1 evidence

Define the H_1 integrand

$$I_1(\Theta_1) = p(d_{GW}|\Theta_1, H_1)p(d_{EM,c}|\Theta_1, H_1)\pi_1(\Theta_1) \tag{C19}$$

so that

$$Z_{1,c} = \int I_1(\Theta_1)d\Theta_1 \tag{C20}$$

For Nnumerical stability, computations are performed in log space:

$$\ell_1(\Theta_1) = \ln(p(d_{GW}|\Theta_1, H_1)) + \ln(p(d_{EM,c}|\Theta_1, H_1) + \ln \pi_1(\Theta_1) \tag{C21}$$

$$Z_{1,c} = \int e^{[\ell_1(\Theta_1)]}d\Theta_1 \tag{C22}$$

The maximum a posteriori point is

$$\hat{\Theta}_1 = \arg\max_{\Theta_1} \ell_1(\Theta_1) \tag{C23}$$

Near $\hat{\Theta}_1$, expand the log integrand to second order. Since the gradient vanishes at an interior maximum, the local form is

$$\ell_1(\Theta_1) \approx \ell_1(\hat{\Theta}_1) - \frac{1}{2}(\Theta_1 - \hat{\Theta}_1)^T A_1(\Theta_1 - \hat{\Theta}_1) \tag{C24}$$

with negative Hessian

$$A_1 = -\nabla\nabla\ell_1(\Theta_1)|_{\hat{\Theta}_1} \tag{C25}$$

Substitutes into Eq. (C22) gives a multivariate Gaussian integral:

$$Z_{1,c} \approx e^{[\ell_1(\hat{\Theta}_1)]}(2\pi)^{\frac{k_1}{2}}|A_1|^{-\frac{1}{2}} \tag{C26}$$

where k_1 is the number of free continuous parameters. In log form,

$$\ln(Z_{1,c}) \approx \ell_1(\hat{\Theta}_1) + (\frac{k_1}{2})\ln(2\pi) - (\frac{1}{2})\ln(|A_1|) \tag{C27}$$

Appendix D Gaussian approximation and Fisher matrix in the consistency statistic

Recall the reduced consistency statistic

$$C_1 = \int \frac{p(\psi|d_{GW}, H_1)p(\psi|d_{EM}, H_1)}{\pi(\psi|H_1)}d\psi \tag{D28}$$

Posterior overlap statistics of this general form are widely used in Bayesian searches for strongly lensed GW event pairs [20], while later lensing frameworks perform more complete joint parameter inference and evidence calculations [21][13].

Assume that the GW and EM distributions are locally well approximated by k_ψ dimensional Gaussian densities,

$$p(\psi|d_{GW}, H_1) \simeq \mathcal{N}(\psi; \mu_{GW}, \Sigma_{GW}) \tag{D29}$$

$$p(\psi|d_{EM}, H_1) \simeq \mathcal{N}(\psi; \mu_{EM}, \Sigma_{EM}) \tag{D30}$$

If the prior varies slowly across the region in which both distributions have appreciable support, it may be approximated locally by a constant $\pi(\psi|H_1) \simeq \pi_0$. The consistency statistic then becomes

$$C_1 \simeq \frac{1}{\pi_0}\int \mathcal{N}(\psi; \mu_{GW}, \Sigma_{GW})\mathcal{N}(\psi; \mu_{EM}, \Sigma_{EM})d\psi \tag{D31}$$

The product integral of two normalised multivariate Gaussian densities is

$$\int \mathcal{N}(\psi; \mu_1, \Sigma_1)\mathcal{N}(\psi; \mu_2, \Sigma_2) = \mathcal{N}(\mu_1; \mu_2, \Sigma_1 + \Sigma_2) \tag{D32}$$

Therefore

$$C_1 \simeq \frac{1}{\pi_0}\mathcal{N}(\mu_{GW}; \mu_{EM}, \Sigma_{GW} + \Sigma_{EM}) \tag{D33}$$

Writing the Gaussian density explitly gives

$$C_1 \simeq \frac{e^{-\frac{D^2}{2}}}{\pi_0 (2\pi)^{\frac{k_\psi}{2}} |\Sigma_{GW} + \Sigma_{EM}|^{\frac{1}{2}}} \tag{D34}$$

with

$$D^2 = (\mu_{GW} - \mu_{EM})^T (\Sigma_{GW} + \Sigma_{EM})^{-1} (\mu_{GW} - \mu_{EM}) \tag{D35}$$

This is the squared Mahalanobis distance [22] between the two posterior means, evaluated usnig their combined covariance.

Fisher information provides a complementary description of measurement precision. For a likelihood $L(d|\psi)$ the expected Fisher matrix is defined by [23]

$$F_{ij} = -\left\langle \frac{\partial^2 \ln L}{\partial \psi_i \partial \psi_j} \right\rangle \tag{D36}$$

under the usual regularity assumptions. Near a well constrined maximum, the log likelihood may be expanded as

$$\ln(L(\psi)) \simeq \ln(L(\hat{\psi})) - \frac{1}{2}(\psi - \hat{\psi})^T F (\psi - \hat{\psi}) \tag{D37}$$

In the asymptotic Gaussian limit, and when prior curvature is negligible,

$$\Sigma \simeq F^{-1} \tag{D38}$$

This approximation is standard in GW parameter estimation, but its domain of validity must be checked carefully: finite SNR ratio, weakly measured parameters, boundaries, degeneracies and informative priors can all make the Fisher approximation inaccurate [24].

For the reduced consistency problem,

$$\Sigma_{GW} \simeq F_{GW}^{-1}, \Sigma_{EM} \simeq F_{EM}^{-1}. \tag{D39}$$

The total covarience becomes

$$\Sigma_{tot} \simeq F_{GW}^{-1} + F_{EM}^{-1}. \tag{D40}$$

Thus the Mahalanobis distance may be written directly in terms of the two information matrices,

$$D^2 = \Delta\mu^T (F_{GW}^{-1} + F_{EM}^{-1})^{-1} \Delta\mu \tag{D41}$$

Substitution into the Gaussian overlap gives.

$$C_1 \simeq \frac{e^{-\frac{1}{2}\Delta\mu^T (F_{GW}^{-1} + F_{EM}^{-1})^{-1} \Delta\mu}}{\pi_0 (2\pi)^{\frac{k_\psi}{2}} |F_{GW}^{-1} + F_{EM}^{-1}|^{\frac{1}{2}}} \tag{D42}$$

Taking the logartithm yields a numerically convenient decomposition,

$$\ln(C_1) \simeq -\ln(\pi_0) - \frac{k_\psi}{2}\ln(2\pi) - \frac{1}{2}\ln(|F_{GW}^{-1} + F_{EM}^{-1}|) - \frac{1}{2}D^2 \tag{D43}$$

The last two terms have distinct interpretations:

1. $-\frac{D^2}{2}$ is the mismatch term. It penalises separation between the GW and EM predictions in units set by the combined uncertainty.
2. $-\frac{1}{2}\ln(|F_{GW}^{-1} + F_{EM}^{-1}|)$ is the precision volume term. It increases as the joint uncertainty volume becomes smaller.

This decomposition makes clear why resolved strong lensing EM information can be highly discriminating under H_1. If the EM lens reconstruction is precise, F_{EM} has large eigenvalues and F_{EM}^{-1} is small. Agreement with the GW posterior can then produce a sharply concentrated overlap. Conversely, a precise EM prediction that is offset from the GW inference can make D^2 large and suppress C_1 exponentially.